$$
\begin{aligned}
\mathrm{Tr}^{o}_{i+k,j+k}(f) = \langle \mathsf{st}, & \mathbf{A}_f + \mathbf{B}_f^{[:,i+k+1:i+k+o]}\mathbf{F}\mathbf{C}_f^{[j+k+1:j+k+o,:]}, \\
& \mathbf{B}_f^{[:,1:i+k]} + \mathbf{B}_f^{[:,i+k+1:i+k+o]}\mathbf{F}\mathbf{D}_f^{[j+k+1:j+k+o,1:i+k]}, \\
& \mathbf{C}_f^{[1:j+k,:]} + \mathbf{D}_f^{[1:j+k,i+k+1:i+k+o]}\mathbf{F}\mathbf{C}_f^{[j+k+1:j+k+o,:]}, \\
& \mathbf{D}_f^{[1:j+k,1:i+k]} + \mathbf{D}_f^{[1:j+k,i+k+1:i+k+o]}\mathbf{F}\mathbf{D}_f^{[j+k+1:j+k+o,1:i+k]}\rangle,
\end{aligned}
$$